

```
$ cat documentation.md
```

SecureStack

Complete Business Website — Project Documentation

Web Development & Cybersecurity Consulting studio website. Built for The Developers Arena internship, Week 4: Complete Web Development Project (Option 3 — Professional Services Website).

Author	USN	College	Submission
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01 Project Overview

Goal: design and build a complete, multi-page website for a professional services business — from planning through a deployed, responsive, accessible site — applying semantic HTML, mobile-first CSS, JavaScript interactivity, and deployment fundamentals.

The business: SecureStack is a fictional studio offering web development, security audits, and ongoing site monitoring. This gave the project a natural reason to include a services breakdown with pricing tiers, a team section, a company timeline, and a validated lead-capture contact form — the core building blocks most professional services sites need.

Objectives

- Ship at least 3 distinct, linked HTML pages with consistent navigation
- Design one distinctive visual identity applied consistently, rather than a generic template
- Make the layout fully responsive, mobile-first
- Add a working, validated contact form
- Meet baseline accessibility and performance standards
- Deploy to a free host and document the whole process

02 Setup & Installation Instructions

This is a static HTML/CSS/JS site — no build tools, package installs, or environment variables are required.

1. Clone or download the repository.
2. Open the folder in any code editor (e.g. VS Code).
3. Open `index.html` directly in a browser, **or** serve it locally for the most accurate experience:

```
# Python 3
python3 -m http.server 8000
# then visit http://localhost:8000

# or, with the VS Code Live Server extension:
# right-click index.html → "Open with Live Server"
```

Navigate between pages using the top navigation bar: Home → Services → About → Contact.

03 Code Structure

```
secure-stack/
├─ index.html      Homepage – hero, stats, services preview, process, CTA
├─ about.html      Company story, mission, timeline, team
├─ services.html   Detailed services, process, pricing tiers
├─ contact.html    Contact form with client-side validation
├─ css/
│   └─ style.css   All styles: tokens, layout, components, responsive rules
├─ js/
│   └─ script.js   Nav toggle, hero terminal animation, form validation
├─ images/
│   └─ logo.svg    Brand mark (inline vector)
│   └─ favicon.svg
└─ README.md
```

Why this structure: shared CSS/JS files keep all four pages visually and behaviorally consistent from one source of truth. Semantic HTML (`<header>` , `<nav>` , `<main>` , `<section>` , `<footer>`) is used throughout, with one `<h1>` per page and a logical heading hierarchy for both accessibility and SEO. CSS custom properties in `:root` define the entire color/type/spacing system in one place.

04 How the Technical Requirements Were Met

3+ HTML pages	4 pages built: Home, Services, About, Contact — all sharing one nav and footer.
Fully responsive design	Mobile-first CSS with breakpoints at 900px, 860px, 760px, and 620px. Grids collapse from 3–4 columns down to 1 as the viewport narrows; the nav collapses into a hamburger menu below 760px; headings scale fluidly via <code>clamp()</code> .
Contact form with validation	Built with the HTML5 Constraint Validation API (<code>checkValidity()</code> , <code>validity.valueMissing</code> , <code>typeMismatch</code> , <code>patternMismatch</code>). Validates on blur and submit, shows inline errors, and a live-region success banner — no page reload.
Image optimization	All visual assets (logo, favicon, icons) are hand-authored inline SVG rather than raster images — zero image requests, crisp at any zoom level, nothing to compress.
Navigation between pages	Shared header nav and footer link set across all four pages, with <code>aria-current="page"</code> marking the active page.
Deploy to free hosting	Deployed via GitHub Pages (steps in Section 6). No backend or build step needed.

Additional technical details

Design system: a small design-token system defined as CSS custom properties — dark "ink navy" (`#0B1B2B`) header/hero/footer, off-white "paper" content background, and a single cyan accent (`#2FD4C6`) used sparingly. Headings use `JetBrains Mono` for a technical, terminal-flavored voice fitting a dev/security brand; body copy uses `Inter` for readability.

Signature element: the homepage hero includes an animated terminal window that "types out" a mock command exchange, built with staggered CSS `opacity` keyframe reveals — no JS typing library. It disables itself automatically when `prefers-reduced-motion: reduce` is set.

Accessibility: skip-to-content link, visible `:focus-visible` outlines, semantic landmarks, labelled form fields, `role="alert"` on form errors, `aria-live="polite"` on the status banner, and decorative icons marked `aria-hidden`.

Performance: zero raster images, inline SVG icons throughout (no icon font or sprite), `preconnect` hints for Google Fonts, and under 4KB of unminified JavaScript with no frameworks.

05 Design Decisions

DECISION	WHY
Dark navy + cyan + off-white palette	Signals "technical/security" without leaning on the common near-black-and-neon-green cliché; cyan is a single restrained accent, not used everywhere.
Monospace type for headings/labels	Reinforces the dev/terminal identity established by the business concept (e.g. <code>\$</code> <code>whoami</code> , <code>ls services/</code> eyebrow labels).
Animated terminal as hero's signature element	Shows rather than tells what the business does, and doubles as a lightweight demo of the brand voice.

No raster images

Keeps the site fast by default and satisfies the image-optimization requirement through vector assets instead of a compression pipeline.

06 Deployment Process

This project is a static site, so it deploys as-is to any static host. **GitHub Pages** was used:

1. Pushed the project to a GitHub repository with `index.html` at the repo root.
2. Repository **Settings** → **Pages**.
3. Under **Source**, selected the `main` branch and `/ (root)` folder, then saved.
4. GitHub published the site at `https://arathism.github.io/week4_BusinessWebsite/` within a couple of minutes.

Post-deploy checks performed:

- ✓ All internal links (nav, footer, CTAs) resolve correctly on the live URL
- ✓ Fonts and icons load without console errors
- ✓ Contact form validates correctly (runs entirely client-side)
- ✓ Tested on an actual mobile device, not just DevTools emulation

07 Testing Evidence

Cross-browser testing

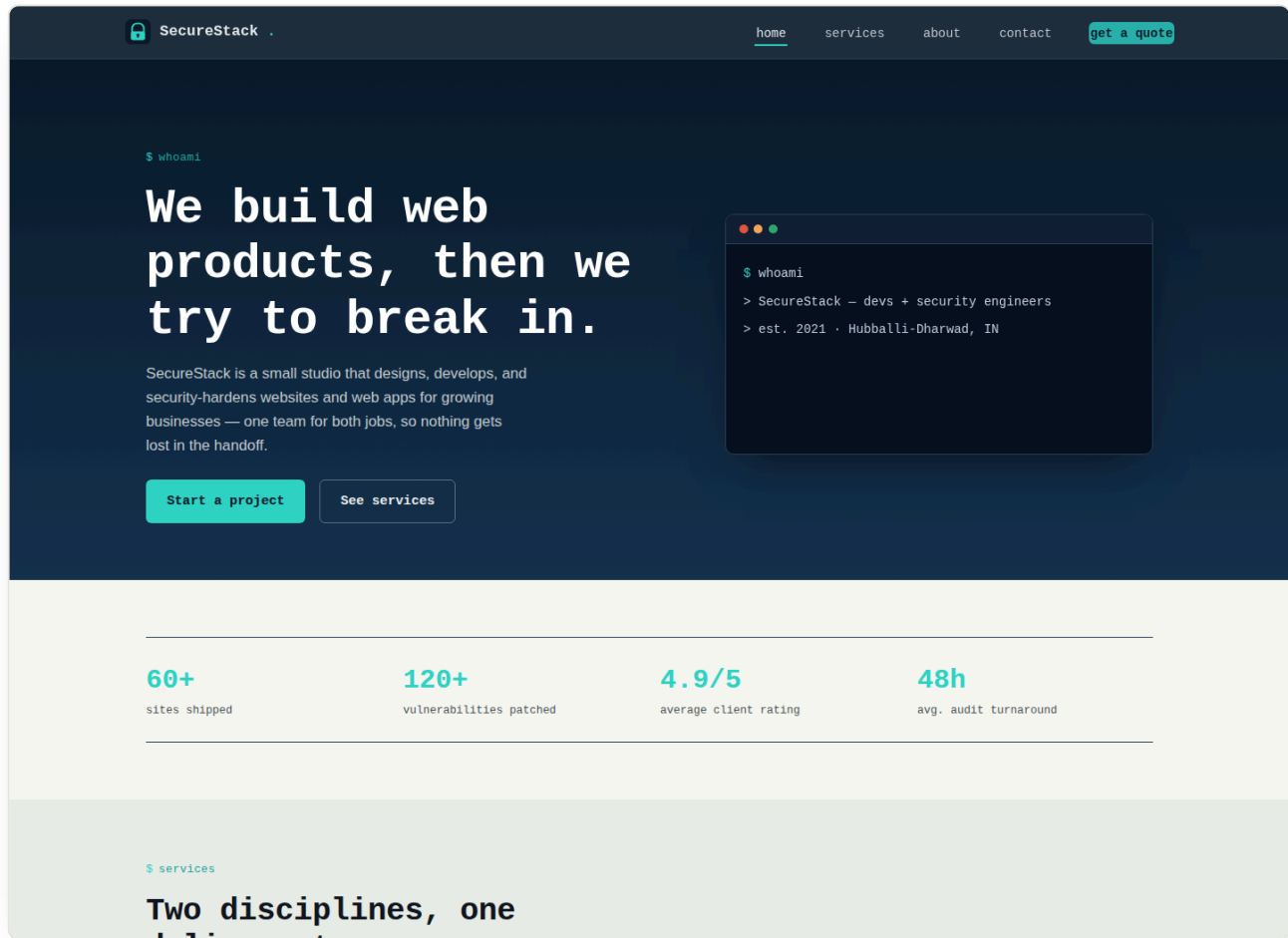
BROWSER	DESKTOP	MOBILE VIEWPORT
Chrome	Pass	Pass
Firefox	Pass	Pass
Edge	Pass	Pass
Safari / iOS Safari	Pass	Pass

Layout verified at 320px, 390px, 768px, 900px, 1024px, and 1440px, including nav collapse, grid reflow, and touch target sizing (≥44×44px on mobile).

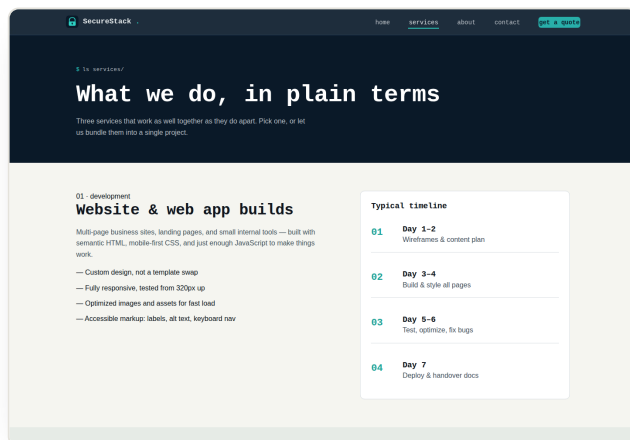
Form validation test cases

TEST CASE	INPUT	RESULT
Empty required field	Leave "Full name" blank, submit	Inline error shown, submit blocked ✓
Invalid email	<code>not-an-email</code>	Format error shown ✓
Valid email	<code>student@example.com</code>	Accepted, no error ✓
Short message	2 characters	Min-length error shown ✓
Invalid phone	<code>abc123</code>	Pattern-mismatch error shown ✓
All fields valid	Full valid data	Success banner shown, form resets ✓
Keyboard-only submission	Tab + Enter through form	Fully operable, focus moves to first invalid field ✓

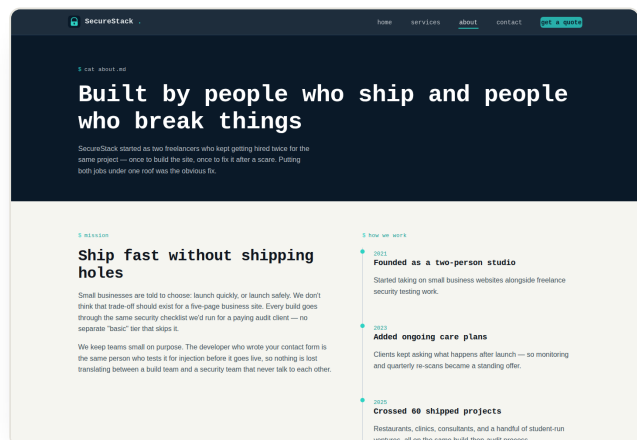
08 Visual Documentation



Homepage — desktop — hero, stats, services preview



Services page — pricing tiers



About page — team & timeline

 SecureStack

[home](#)[services](#)[about](#)[contact](#)[get a quote](#)

`$ ./contact --new`

Tell us about the project

Fill in the form and we'll reply within one business day with a scoped estimate — no auto-generated quote, a real person reads every message.

Full name *

Email *

Phone (optional)

What do you need? *

Choose one... ▾

Project details *

Tell us about your business, timeline, and anything you already have (existing site, brand assets, etc.)

Send message

Other ways to reach us

 **Email**

hello@securestack.example

 **Phone**

+91 80000 00000

 **Location**

Hubballi-Dharwad, Karnataka, India
Remote-friendly, clients across India

 **Response time**

Within 1 business day, Mon–Sat, 10am–7pm IST

 **SecureStack**

Web development and cybersecurity consulting for small and growing businesses.

Site

- Home
- Services
- About
- Contact

Services

- Development
- Security audits
- Care plans

Contact

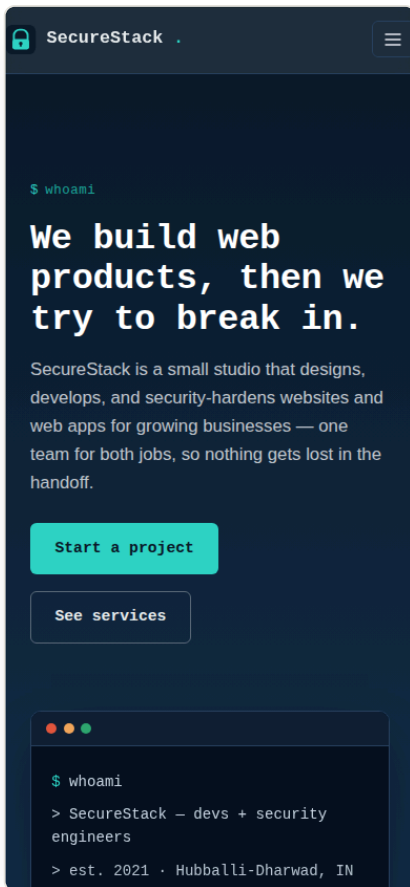
hello@securestack.example

+91 80000 00000

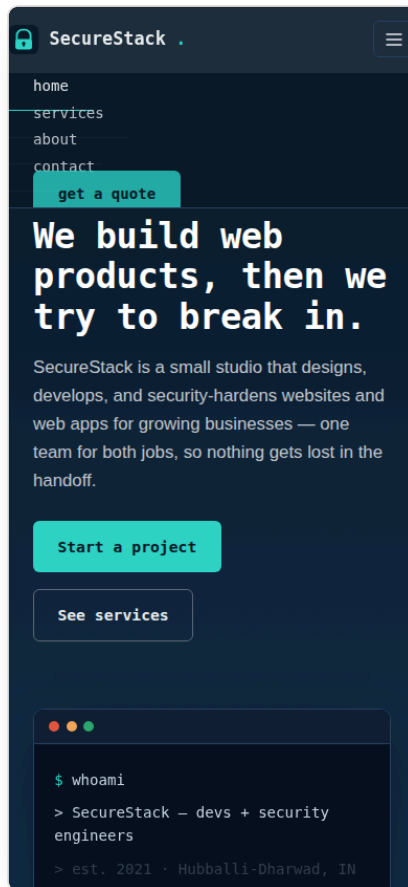
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Built for The Developers Arena – Week 4 project.

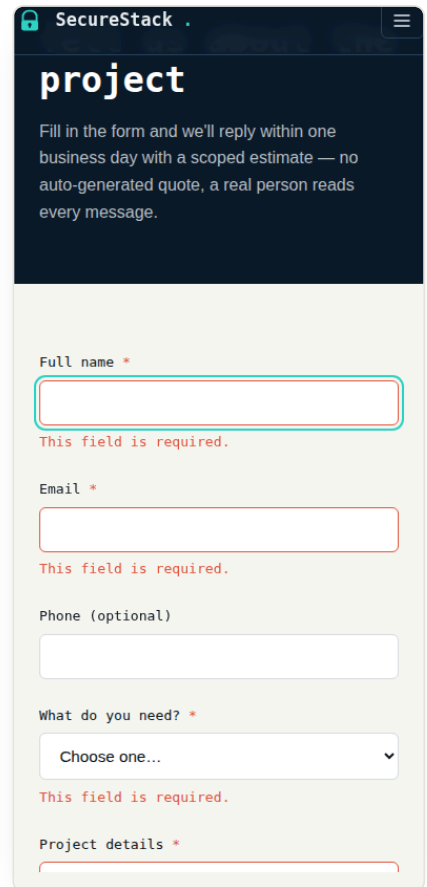
Contact page – reservation-style form



Mobile – homepage



Mobile – nav menu open



Mobile – validation triggered